

An Overview of Force Measurement

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Abstract—The abstract goes here.

Index Terms—Force transducer, strain gauge, electrical methods, measurements, elastic element, Wheatstone bridge,

I. INTRODUCTION

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II. FORCE SENSING METHODES

There are many methods to measure forces exerted on objects. These methods differ from one another in the physical principle of sensing. In this section, we will explain some of these methods and compare between advantages and disadvantages of each method As shown in Table (??).

- A. Resistive force transducer
- B. Capacitive force transducer
- C. Inductive force transducer
- D. Piezoelectric force transducer
- E. Fiber optic force transducer

There are various optical methods for measuring force [1]. One of these methods includes monitoring the analog optical signal output of an optical LED [2]. Another method involves connecting an LED to a camera and using computer vision algorithms to measure the force. In this section, we examine force measurement using Fiber Bragg Grating (FBG) technology. It is a device that converts mechanical Force or strain into a change in light transmission properties. A Bragg fiber consists of a transparent fiber containing a large number of reflection points distributed at periodic intervals as shown in Fig. (1).

The wavelength of the light reflected from the Bragg grating depends on the periodic spacing between these points as shown in relationship (1).

$$\lambda = 2n_{eff}\Lambda \quad (1)$$

where n_{eff} is the effective refractive index of the fiber core and λ is the wavelength of the light reflected by the Bragg grating, Λ grating period. When subject to mechanical force, the spacing of reflection points changes, resulting in a change

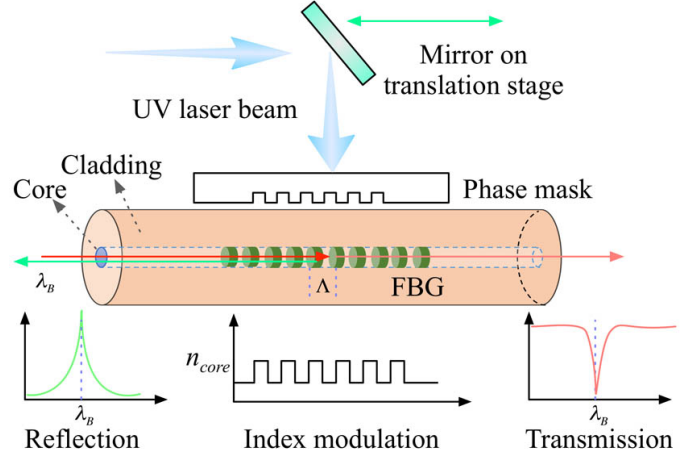


Fig. 1: Schematic of FBG imprinted in an SMF by the phase-mask technique with a scanning UV laser beam [3].

in the reflected light wavelength. This change is proportional to the induced force or strain as shown in (2) [4].

$$\frac{\Delta\lambda}{\lambda} = k\varepsilon \quad (2)$$

where λ is wavelength of the Bragg grating, $\Delta\lambda$ wavelength change at the force or strain impressed in the grid, k gauge factor, ε strain. Bragg grating fibers function as filters, transmitting certain wavelengths while reflecting the rest. The gratings within these optical fibers are created using an intense UV source such as a UV laser [5]. It is possible to use photosensitive like germanium-doped silica fibers [5] which means that the refractive index of the core changes when exposed to UV light. Bragg grating fiber can be fabricated to monitor force at multiple points by manufacturing the fiber with varying grating periods (L) in distinct sections [5]. Fiber Bragg grating-based force sensors require precise alignment and careful placement to achieve accurate measurements. Processing the output signal from these gratings necessitates the use of external interrogation equipment, which poses a challenge for mobile robotics applications [1]. In the literature, Wang et al. [6] present a miniature triaxial force sensor based on fiber Bragg grating (FBG) for detecting the interaction forces between the tip of flexible endoscopic surgical robots and tissue. The prototyped sensor has been calibrated and achieves a high resolution. Loading experiments results show that the sensor can provide a high-accuracy measurement of three force components with an average RMS error of less than 3% in the total operating ranges. Xiong et al. [7] present three-dimensional force sensor based on fiber Bragg grating (FBG) for robot planar force measuring. Three-dimensional forces of F_x , F_y , and F_z have been measured by only five sensitive elements of FBG experimental results demonstrate that the

sensor possesses good linearity, weak coupling and creep resistance. Dong et al. [8] conducted a review of force sensors based on fiber Bragg gratings used in minimally invasive surgery and robot-assisted minimally invasive surgery. They further review force/temperature decoupling approaches from calibration-matrix methods to learning-based models. Furthermore, they outlined research avenues to enhance robustness and validate efficacy, paving the way for the development of clinically applicable forced-feedback visual systems.

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F. Vibrating wire force transducer

III. STATIC VS DYNAMIC FORCE MEASUREMENT

IV. MULTI-DIMENSIONAL FORCE MEASUREMENT

V. FORCE ESTIMATION

VI. FORCE MEASUREMENT APPLICATIONS

A. Force measurement in monitoring and testing tasks

B. Use of force measurement in automated systems

C. Force measurement in research and development laboratories

VII. CONCLUSION

The conclusion goes here.

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